

Question 1

Given below is a longitudinal section of a bean seed. Label the parts marked 1 to 5 and write their functions.

Testa — It protects the seed from insect and mechanical injury.

Plumule — It develops into the shoot.

Radicle — It develops into the root.

Micropyle — It absorbs and allows the entry of as much water as is for germination.

Cotyledons — stored food material which is used by the seedlings for growth.

Question 2

Name the following:

- | | |
|---------------|-----------------|
| (a) Pea seed | (b) Maize grain |
| (c) Bean seed | (d) Papaya seed |

Question 3(a)

Differentiate between Radicle and Plumule

The radicle grows downward. The plumule grows upward.

The radicle develops into root. The plumule develops into shoot.

Question 3(b)

Differentiate between Hilum and Micropyle

Answer

Differences between Hilum and Micropyle are:

Hilum	Micropyle
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Hilum is a scar which is present on the inner concave side of the seed. Micropyle is a small pore which is present above the hilum.

Hilum is the place where the seed was attached to the fruit wall. Micropyle absorbs and allows the entry of as much water as is required for germination.

Question 3(c)

Differentiate between Testa and tegmen

Answer

Differences between Testa and Tegmen are:

Testa	Tegmen
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Testa is the protective outer covering of the seed coat.

Tegmen is the protective inner part of the seed coat.

Testa is smooth, thicker and impermeable. Tegmen is thin and membranous.

Testa protects the seed from insect, bacteria as well as mechanical injury. Tegmen protects the embryo from dehydration and mechanical damages.

Question 4

Give two functions of a fruit.

Answer

Two main functions of a fruit are:

- Protects seed from the unfavourable environmental condition.
- Fruit stores food inside them.

Question 5

Radicle emerges out of the seed earlier than plumule. State one advantage served by this.

Answer - The Radicle emerges out of the seed earlier than the plumule and absorbs water and essential nutrients from the soil.

- This helps the plumule to grow and allows for photosynthesis.

Question 7

(a) Radicle - develops into root.

(b) Cotyledons - contain stored food material which is used by the seedling for growth.

(c) Endosperm - stores food in the form of starch.

(d) Micropyle - absorbs and allows the entry of as much water as is needed for germination.

Question 9

Name the part of the seed from which the following are

- (a) Radicle
- (b) Plumule

Question 11 Define the terms given below:

(i) Fruit — A fruit is the ripened ovary. It is a sweet, fleshy, edible plant part.

(ii) Seed — A seed is a fertilized ovule, from which a new plant of the same kind can grow.

(iii) Fertilization — The fusion of male and female sex cells (gametes) is called fertilization.

(iv) Hilum — Hilum is a scar on the inner concave side of the seed, which marks the place where the seed was attached to the fruit wall.

(v) Dry fruit — A dry fruit is the one in which the pericarp is not pulpy and encloses the seeds. For example, the pea pod.

Long Answer Questions

Question 1 What is meant by pollination? Name the two types of pollination.

ANSWER - Pollination is the transfer of pollen grains from the anthers to the stigma of a flower. The two types of pollination are:

1. Self pollination

2. Cross pollination

Question 2 Imagine all the seeds produced by a plant happen to fall under the same plant and sprout into new plants. Mention any two problems that will be faced by the new plants.

Answer

The two main problems that will be faced by the new plant are:
New plants will lack space for their growth because comparatively large number of plants are growing at a particular place.

The new plants will face competition among themselves to get nutrient from the soil which will hamper their growth and development.

Question 3

What is a flower? Draw a typical flower and label its different parts.

Answer

Flower is a specialized shoot in which the leaves are modified into floral structures. It is the reproductive organ of the plant.

Question 4

With the help of a suitable labelled diagram, describe the structure of a dicot seed.

Answer - Taking Bean seed as an example of a dicot seed, below is its labelled diagram:

The structure of a bean seed is described below:

The bean seed is protected by a thin, greenish outermost covering called seed coat. It protects the seed from insects and bacteria as well as from mechanical injury.

The seed coat is divided into two parts, the outer part is called Testa and inner part is called Tegmen. Tegmen is a thin membrane that lies under the Testa.

On the inner concave side of the seed, there is a scar called Hilum which marks the place where the seed was attached to the fruit wall.

Above the hilum is a small pore called Micropyle. The Micropyle absorbs and allows the entry of as much water as is required for germination.

On removing the testa and tegmen from a soaked bean seed, the seed is made up of two fleshy seed leaves called

the Cotyledons. They contain stored food material which is used by the seedling for growth.

In between the two cotyledons is located the delicate embryo which consist of a Radicle and Plumule. The radical develops into a root, while the plumule develops into a shoot.

Question 5

Define germination? Name the two types of germination. Explain with examples.

Answer

The process by which an embryo within the seed becomes active and grows into a young plant is called germination.

The two types of germinations are:

Epigeal Germination — Put some bean seeds in moist cotton placed in a petri dish. After about two days, the radicle grows downwards and develops the roots, while the plumule grows upwards and develops into the shoot. It is seen that the cotyledons are pulled above the surface of the ground. This type of germination in which the cotyledons are pushed above the soil is called Epigeal Germination.

Hypogeal Germination - Soak some maize grains in water for about 24 hours. The stored food in the endosperm is used by the radicle and plumule for their growth. The radicle grows downward while the plumule grows upward and forms the first leaf. This type of germination is called hypogeal germination, as the cotyledons remain under the soil. The plumule only comes out of the soil to form leaves of the plant.

Question 6

What are the three conditions necessary for the germination of seeds?

Answer

For the germination of the seeds the three necessary conditions are:

Water — All living things require water to carry out their life processes. Most chemical reactions occur in water. So seeds also need water during germination.

Oxygen — Seed also needs air for the germination of seed. The oxygen present in the air is used to oxidize the stored food and thus release energy. The energy is required for the growth of the embryo.

A Suitable Temperature — It is necessary for all chemical activities assisted by enzymes. Enzymes are inactive at low temperatures and get destroyed at higher temperatures. They act best between 35° and 40°C.

Question 7

Give the main differences between Hypogeal and Epigeal germination.

Answer

Differences between Hypogeal and Epigeal germination are:

Hypogeal	Epigeal
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The Cotyledons remain under the soil.	The Cotyledons are pushed above the soil.
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Epicotyl elongates faster than Hypocotyl, hence cotyledons remain below.	Hypocotyl elongates faster than Epicotyl, hence cotyledons get pulled above.
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Example - Maize, Rice.	Example - Papaya, Bean.
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Question 8

State the location of the following in a flower:

(a) Sepals

(b) Petals

(c) Anthers

(d) Stigma

Answer

(a) Sepals — Sepals are located in the Calyx which is the outermost whorl of the flower.

(b) Petals — Petals are located in the Corolla which is the second inner whorl of the flower.

(c) Anthers — Anthers are located at the tip of the stamens in the third whorl of the flower, inner to the petals.

(d) Stigma — Stigma is located in the fourth and innermost whorl of the flower, in the pistil at the tip of the style.

Question 9(a)

Give the difference in function between Ovary and Ovule.

Answer

Difference in function between Ovary and Ovule is:

Ovary Ovule

Ovary after fertilization turns into a fruit. Ovule after fertilization turns into a seed of a fruit.

Question 9(b)

Give the difference in function between Petal and Sepal.

Answer

Difference in function between Petal and Sepal is:

Petal Sepal

The petal makes the flower attractive and hence attract insects for pollination. The sepal encloses the inner parts of the flower to provide necessary protection to the growing bud.

Question 9(c)

Give the difference in function between Filament and Style.

Answer

Difference in function between Filament and Style are:

Filament Style

The filament bears anther at its tip which produce pollen grains. Pollen grains contain the male gametes. The style transfers the male gametes of the pollen grains into the ovary.

Filament comes under the male reproductive part of the flower. Style comes under the female reproductive part of the flower.

Question 9(d)

Give the difference in function between pollen grains and ovule.

Answer

Difference in function between pollen grains and ovule are:

Pollen grains	Ovule
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It contains the male gametes.	It contains the female gametes.
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Pollen grains deliver male gametes from stamen of the flower to ovule. Ovule comes under the female reproductive part of the flower.

Question 10

Draw a neat diagram of the 'longitudinal section of a maize grain' and label its seven important parts. Answer the following questions :

(i) Name the structure which divides the embryonic and endospermic region of the grain.

(ii) Among the parts labelled by you above, which one of them is protienaceous in nature ?

(iii) Name the food component which is stored in the endosperm.

(iv) Mention the part of the seed which develops into the root.

(v) Why do we call it as a 'grain' and not as a seed or fruit ?

Answer

(i) The epithelial layer divides the embryonic and endospermic region of the grain.

(ii) Aleurone layer is protinaceous in nature.

(iii) Endosperm stores food in the form of starch.

(iv) Radicle is the part of the seed which develops into the root.

(v) We call it 'maize grain' as the seed coat is fused together with the pericarp, making it indistinguishable.

Question 11

Given alongside is a figure of the vertical section of a typical bisexual flower. Study the same and answer the following questions :

(i) Label the guidelines 1 to 9.

(ii) Which of the above labelled parts constitute the pistil? Write the numbers and names of those parts.

(iii) Name the part which produces pollen grains.

(iv) Is it a complete/incomplete flower ? Give reason in support of your answer.

(v) Write one important role of the part labelled as 6.

Answer

(i) The parts 1 to 9 given in the diagram are:

1 → Thalamus

2 → Sepal

3 → Filament

4 → Anther

5 → Petal

6 → Stigma

7 → Anther

8 → Style

9 → Ovule

(ii) The parts in the given figure that constitute the pistil are:

6 → Stigma,

8 → Style, and

9 → Ovules.

(iii) The anther contains pollen sacs which produce a yellow, powdery material consisting of pollen grains.

(iv) The given figure shows a complete flower. This is because all four whorls viz. calyx, corolla, androecium and gynoecium are present in the same flower.

(v) One important role of Stigma is that it receives the pollen grains during pollination.

Question 12

The figure given alongside represents the vertical section of a fruit. Study the same and answer the following questions :

(i) Identify the fruit and mention whether it is a true fruit or a false fruit.

(ii) Label the guidelines 1, 2, 3 and 4.

(iii) Write the collective name for the parts shown as 1, 2 and 3.

(iv) Name the part of the flower which matures into part 4.

(v) Which of the above labelled parts of the fruit is fleshy and edible ?

Answer

(i) The given fruit is Mango. Mango is a true fruit as the fleshy part and the seed both develop from the ovary.

(ii) The parts 1 to 4 are:

1 → Epicarp

2 → Mesocarp

3 → Endocarp

4 → Seed

(iii) Epicarp, Mesocarp, and Endocarp are collectively known as the Pericarp.

(iv) The ovule that contains the fertilised cell develops into a seed.

(v) Mesocarp is the sweet, fleshy and edible layer.

Case Study Type Questions

Question 1

One winter morning, Shweta and her mother were sitting in their garden when they noticed fresh pea flowers blooming.

Over the next few days, they watched some of these flowers slowly develop into tiny green pea pods. Together, they began tracking each stage of this change and observed how pollination and fertilization occur in plants. Answer the following questions:

(i) What role do insects play in the development of a flower into a fruit?

(ii) Mention the male and female gametes of the flower.

(iii) What is the function of the pollen tube during fertilization?

(iv) What happens to the ovary once fertilization is completed?

(v) Shweta's mother claimed that the pea pod is a fruit and not a vegetable. Explain.

Answer

(i) Insects act as pollinating agents. When the insect visits a flower to collect nectar, the pollen grains stick to its

mouth-parts, wings, legs, etc. When this insect visits another flower of the same kind, the pollen grains from its body may fall on its stigma and bring about fertilization. Transfer of pollen by insects is called insect-pollination.

(ii) The androecium is called the male reproductive part of the flower. The gynoecium is called the female reproductive part of the flower.

(iii) Pollen grains germinate to produce pollen tubes. One of the pollen tubes continue to develop downwards into the style. This pollen tube carries the male gametes. The pollen tube finally reaches the ovary.

(iv) The ovary after fertilization becomes larger and develops into a fruit. The ovule containing the fertilized cell develops into a seed.

(v) A pea pod is a fruit because it develops from the ovary of the flower after fertilization and contains seeds.